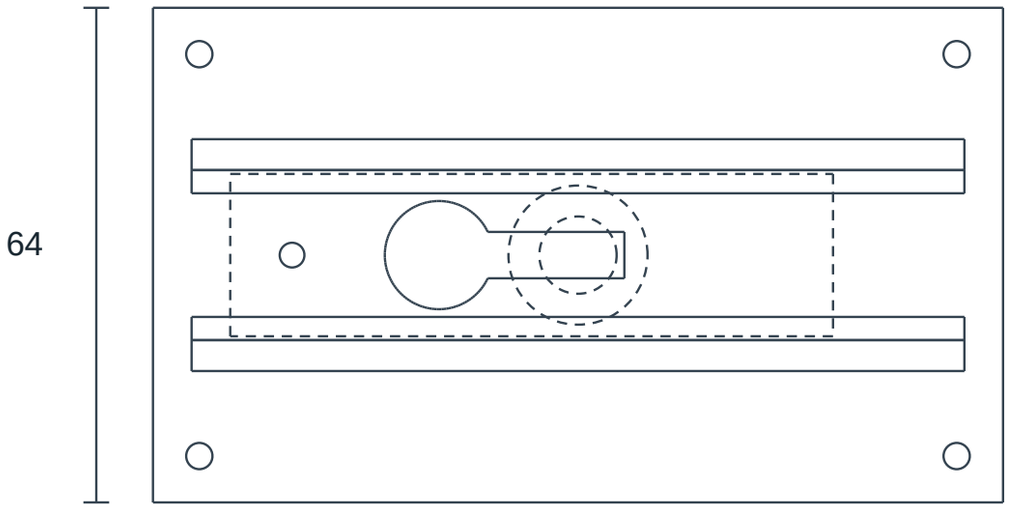
R01 / independent keyhole gate bench reference

REFERENCE DESIGN (NEW) - unbuilt; not historical SkyRelief geometry

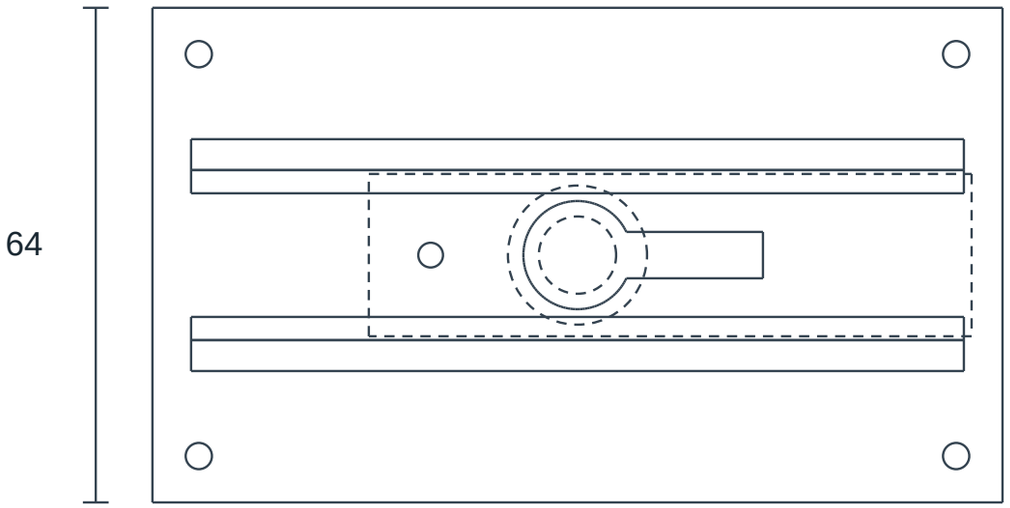
Units: mm. Coordinates: +X gate travel, +Y in base plane, +Z upward. No material/load/servo specification.

LOCKED / gate at X = 10



Solid: chassis/rails and aperture; dashed: gate/window/coupon

RELEASED / gate at X = 28



Solid: chassis/rails and aperture; dashed: gate/window/coupon

NEW DESIGN TABLE

Base 110 x 64 x 5; four mounting holes diameter 3.4 at X=6/104, Y=6/58.

Gate 78 x 21 x 3 at Z=5.5 to 8.5; rail lips Z=9 to 11; nominal top/bottom/lateral gap 0.5.

Gate aperture diameter 14; neck width 6; coupon flange diameter 10 x 2, stem diameter 5.

Chassis opening diameter 18 at (55,32); release aperture translates from X=37 to X=55.

Actuator attachment hole diameter 3.2 at locked (18,32); no linkage, actuator or stop is designed.

Reference drawing is nominal only: no manufacturing allowance, tolerance, load rating or fit qualification.

Digital BRep and 19 sampled translation checks do not establish continuous motion or powered operation.

Source: cad/independently_authored/build_reference.py and parameters.json | 0.1.0-review | 2026-10-20